Mahan Choudhury

Hamilton, ON | +1 (289) 933-8844 | mahanc2210@gmail.com | linkedin.com/in/mahan-choudhury-989882192

Available for Co-op / Internship, from May 2027 to April 2028

Summary

MASc student in Interdisciplinary Biomedical Engineering at McMaster University, focused on computer vision and deep learning. I adapt Vision Transformer foundation models to heterogeneous real-world image data, detect and solve drift cases in medical imaging and image-quality problems that degrade model performance, and build GPU-accelerated image pipelines on HPC clusters. First-author journal publication in explainable deep learning.

Education

McMaster University — Hamilton, ON, Canada
MASc in Biomedical Engineering — *Sep 2025 to Present*
Coursework: Machine Learning, Deep Learning, Speech Recognition, Statistical Methods in Health Research

Bangladesh University of Professionals — Dhaka, Bangladesh
B.Sc. in Information and Communication Engineering, CGPA 3.89/4.00 — *Jan 2020 to Dec 2023*
Coursework: Artificial Intelligence, Machine Learning, Data Warehouse and Data Mining, Cloud Computing, Software Engineering

Technical Skills

Languages:	Python, C, SQL, Java, JavaScript, PHP
ML / DL:	PyTorch, TensorFlow/Keras, scikit-learn, timm, NumPy, pandas, LoRA fine-tuning, transfer learning
Computer Vision:	Vision Transformers (ViT, DINOv2/v3), CNNs (GoogLeNet, EfficientNet, ResNet), CLIP-style contrastive models, U-Net segmentation, image preprocessing
GPU / HPC:	CUDA-enabled PyTorch, CuPy, multi-GPU (Tesla V100) clusters, SLURM, Linux, SSH, Github
Data / Other:	MySQL, SPSS, open-weight LLMs (Qwen), Jupyter, LaTeX, Agile/Scrum
Cloud Experience:	Azure Virtual Machines

Research Experience

Graduate Researcher, McMaster University — Sep 2025 to Present
Supervisor: Dr. Ashirbani Saha — *Hamilton, ON*

- Developing few-shot LoRA adaptation of a DINOv3 Vision Transformer for view-invariant image classification on an ordinal scale (BI-RADS breast density), across three public mammography datasets.
- Designed a dual-stream model fusing the LoRA-adapted ViT with a frozen EfficientNet-B5 encoder, using per-stream L2 normalization and temperature-scaled cosine prototypical classification; evaluated with weighted kappa, MAE and adjacent accuracy.
- Built a distribution shift detection framework on frozen ViT embeddings across three clinical sources, reporting results at image, laterality and patient level with site-stratified error analysis and UMAP visualization.
- Explored physics-informed synthetic image degradations to stress-test model robustness.
- Wrote GPU-accelerated preprocessing (PyTorch tensor ops, CuPy connected-component analysis) for four large datasets at 2048×1536 resolution.
- Diagnosed and fixed model and pipeline failures: an input-resolution mismatch degrading a pretrained encoder, checkpoint key mismatches between libraries, a PyTorch 2.6 model-loading change, and a faulty GPU node on the SLURM cluster.

Applied AI Project: LLM-Assisted Systematic Review Screening (PEBC) — 2026 to Present
McMaster University — *Hamilton, ON*

- Building a pipeline that runs open-weight LLMs (e.g., Qwen) on a SLURM GPU cluster to screen literature records against inclusion and exclusion criteria (title/abstract, then full text) for clinical guideline developers.
- Developing a locally hosted interface to compare screening results across models.

Undergraduate Thesis, Bangladesh University of Professionals — Jan 2023 to Dec 2023
Supervisor: Dr. Mohammad Abu Yousuf — *Dhaka, Bangladesh*

- Detected sleep apnea from single-lead ECG by converting band-pass filtered signals to scalograms (Continuous Wavelet Transform) and benchmarking several pretrained CNNs with transfer learning.
- Applied LIME to explain model decisions on scalogram images; work published as a first-author journal paper.

Publication

M. Choudhury, M. Tanvir, M. A. Yousuf, N. Islam, and M. Z. Uddin, "Explainable AI-driven scalogram analysis and optimized transfer learning for sleep apnea detection with single-lead electrocardiograms," *Computers in Biology and Medicine*, vol. 187, p. 109769, Feb. 2025. doi:10.1016/j.compbiomed.2025.109769

Selected Projects

- **Vision Transformer from Scratch (PyTorch):** implemented ViT ("An Image Is Worth 16x16 Words"), including patch embedding, multi-head self-attention, positional encoding and the classification head.
- **Multimodal Vision-Language Model (PyTorch):** combined a SigLIP-style ViT with a Transformer language backbone using contrastive learning; resolved numerical stability issues in softmax and cross-entropy.
- **Biomedical Image Segmentation with U-Net (PyTorch):** implemented and trained a U-Net with encoder-decoder skip connections and Dice-based evaluation.
- **Transformer from Scratch (PyTorch):** implemented the encoder-decoder Transformer with scaled dot-product attention, multi-head attention, positional encoding and masking.
- **Cardiovascular Disease Detection (scikit-learn):** majority-vote ensemble of five classifiers; Random Forest led across three datasets.
- **Web-App:** Hospital Management System (Laravel, MySQL), Blood Bank Management System (Java, MySQL), Power BI sales dashboard.

Work Experience

Teaching Assistant, IBEHS 4D03: Introduction to Medical Imaging Jan 2026 to Present
McMaster University *Hamilton, ON*

- Coordinate weekly labs on MRI, NMR and DTI; develop Python and Jupyter teaching materials; grade lab reports against rubrics; give one-on-one debugging support for image processing pipelines.

Web Development Intern, Business Automation Limited Oct 2022 to Dec 2022
ICT Division project *Dhaka, Bangladesh*

- Worked with cross-functional teams on a client-facing Laravel application, contributing to dashboard development and large-scale data management.

Awards and Leadership

- **Vector Scholarship in Artificial Intelligence** (2025-26), merit-based, valued at $17,500
- BUP Scholarship for academic performance (top 3%); Dean's Appreciation Letter for a 4.00/4.00 semester GPA
- Symposium Officer, BMEGA (2025 to 2026); Vice President (Operations), BUP Robotics Club (2023 to 2024); Community Developer, Microsoft Learn Student Ambassadors, Git workshops (2021 to 2022)